DIVIT
ELECTRONICS

# Acceptance Test Procedure

## SH2000 Dual-Ridge Horn Antenna, 2–32 GHz

| | |
|---|---|
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| **Stock no** | 5985-01-234-5678 |
| **Revision** | B |
| **Release date** | 07.11.2026 |
| **Classification** | Company confidential |

| Prepared by | Checked by | Approved by | Release |
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| Signature: | Signature: | Signature: | Transfer date: 07.11.2026 |

# Document change record

*Table 1 – Document change record*

| Rev. | Date | Description | Prepared | Approved |
|---|---|---|---|---|
| A | 12.05.2026 | First issue | E. Kaya | M. Demir |
| B | 07.11.2026 | Software installation steps and record form added | E. Kaya | M. Demir |



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# 1 Scope

This procedure describes how the SH2000 dual-ridge horn antenna is tested before it is accepted: the hardware and software needed, how the test set-up is built, how the software is prepared and installed, and the steps of each test. Measured values are recorded on the form in Annex A, not in this document.

The procedure applies to every SH2000 delivered under the project, at the acceptance test after production.

# 2 General information

## 2.1 Project

The SH2000 is used as the reference antenna of the **Divit EMC Range** project, where it measures the gain and pattern of the antennas under development.

## 2.2 Stock definition

*Table 2 – Stock definition*

| Item | Value |
|---|---|
| Stock no | 5985-01-234-5678 |
| Part no | 617-355-SH2000 |
| Manufacturer | Microwave Vision Group |
| Description | Dual-ridge horn antenna, 2–32 GHz |
| Unit of issue | Each |

## 2.3 Unit introduction

The SH2000 is a broadband dual-ridge horn. The ridges keep the impedance near 50 Ω over the whole band, so one antenna covers 2–32 GHz. It is fed through an SMA connector at the back of the waveguide section.



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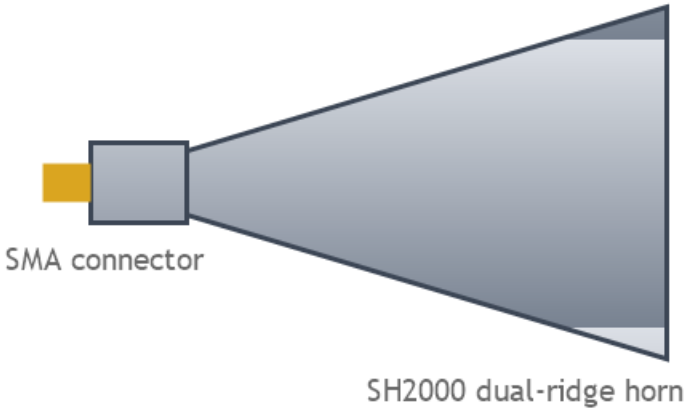



*Figure 1 – SH2000 dual-ridge horn antenna*

*Table 3 – Main characteristics*

| Characteristic | Requirement | Verified in |
|---|---|---|
| Frequency range | 2–32 GHz | 5.1 |
| VSWR | ≤ 2.0 : 1 | 5.1 |
| Gain | 3–17 dBi | 5.2 |
| Mass | ≤ 0.3 kg | Visual inspection |

# 3 Documents

## 3.1 Applicable documents

The following documents form part of this procedure; their latest revision applies.

*Table 4 – Applicable documents*

| No | Document no | Title | Rev. |
|---|---|---|---|
| AD1 | SPEC-SH2000-01 | SH2000 product specification | C |
| AD2 | QP-TEST-004 | Calibration of test equipment | F |
| AD3 | IEEE Std 149-2021 | Recommended practice for antenna measurements | - |

## 3.2 Reference documents

*Table 5 – Reference documents*

| No | Document no | Title | Rev. |
|---|---|---|---|
| RD1 | UM-VNA-220 | Vector network analyser user manual | 3 |



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# 4 Abbreviations and symbols

*Table 6 - Abbreviations*

| Abbreviation | Meaning |
|---|---|
| ATP | Acceptance test procedure |
| AUT | Antenna under test |
| DUT | Device under test |
| SMA | SubMiniature version A connector |
| VNA | Vector network analyser |
| VSWR | Voltage standing wave ratio |

*Table 7 - Symbols*

| Symbol | Quantity | Unit |
|---|---|---|
| f | Frequency | GHz |
| G | Gain | dBi |
| Γ | Reflection coefficient | - |
| $\lambda_0$ | Free-space wavelength | m |

# 5 Tests

## 5.1 Return loss (VSWR) test

The test checks that the antenna is matched to 50 Ω over 2–32 GHz: the VSWR may not exceed 2.0 : 1 at any frequency.

### 5.1.1 Hardware list

*Table 8 - Hardware list*

| No | Equipment | Manufacturer | Model | Serial no | Calibration due |
|---|---|---|---|---|---|
| 1 | Vector network analyser | Rohde & Schwarz | ZNA43 | 101234 | 03.2027 |
| 2 | Calibration kit, 2.92 mm | Rohde & Schwarz | ZN-Z229 | 100987 | 01.2027 |
| 3 | Test cable, 1 m | Huber+Suhner | SF102 | C-0441 | - |
| 4 | Torque wrench, 0.9 N·m | Rohde & Schwarz | ZN-ZTW | T-017 | 06.2027 |
| 5 | Absorber panel | Divit | ABS-60 | - | - |



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### 5.1.2 Software list

*Table 9 - Software list*

| No | Software | Version | Purpose |
|---|---|---|---|
| 1 | VNA firmware | 2.40 | Runs the network analyser |
| 2 | Divit Antenna Test Suite | 1.3.0 | Drives the measurement and writes the record file |
| 3 | Calibration kit data | ZN-Z229 V2 | Standards of the calibration kit |

### 5.1.3 Test set-up

The antenna points at the absorber panel at least 1 m away, so the reflections of the room stay below the measured return loss (Figure 2).

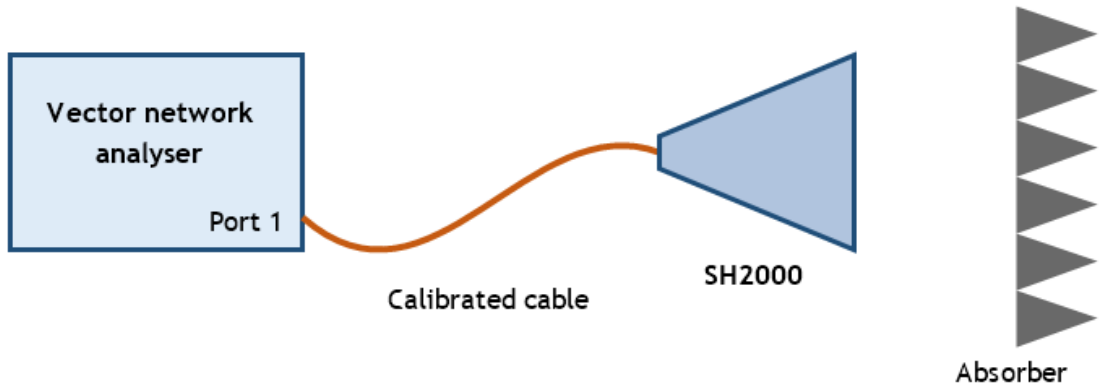



*Figure 2 - Return loss test set-up*

### 5.1.4 Software preparation and installation

**Caution:** install the software with the analyser switched off; a running analyser may lose its calibration.

*Table 10 - Software installation steps*

| Step | Action | Expected result | ✓ |
|---|---|---|---|
| 1 | Copy the Divit Antenna Test Suite 1.3.0 installer to the test computer. | Installer present | ☐ |
| 2 | Run the installer as administrator and accept the default folder. | Installation finished without error | ☐ |
| 3 | Load the ZN-Z229 calibration kit data into the analyser. | Kit listed under Calibration > Kits | ☐ |
| 4 | Start the test suite and select the SH2000 test plan. | Plan opens, 2–32 GHz, 1601 points | ☐ |



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### 5.1.5 Test steps

*Table 11 – Return loss test steps*

| Step | Action | Expected result | ✓ |
|---|---|---|---|
| 1 | Switch the analyser on and let it warm up for 30 minutes. | Ready indication | ☐ |
| 2 | Calibrate port 1 at the end of the test cable (open, short, load). | Calibration valid | ☐ |
| 3 | Connect the SH2000 with the torque wrench at 0.9 N·m. | Connector tight, not over-torqued | ☐ |
| 4 | Start the sweep from the test suite. | VSWR curve shown | ☐ |
| 5 | Read the highest VSWR and record it on form A-1. | VSWR ≤ 2.0 : 1 | ☐ |

## 5.2 Radiation pattern test

The pattern is measured in the anechoic chamber with the positioner; the hardware, set-up and steps follow the same layout as 5.1 and are given in the chamber procedure (RD1).



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# 6 Annexes

## 6.1 Annex A – Test record form

Fill in one form per antenna; the form is kept with the delivery documents.

*Table 12 – Form A-1: return loss test record*

| **Antenna serial no** | | **Test date** | |
|---|---|---|---|
| Tested by | | Test station | |
| Results | | | |
| Frequency band | Highest VSWR | Limit | Result |
| 2–8 GHz | | ≤ 2.0 : 1 | ☐ Pass ☐ Fail |
| 8–18 GHz | | ≤ 2.0 : 1 | ☐ Pass ☐ Fail |
| 18–32 GHz | | ≤ 2.0 : 1 | ☐ Pass ☐ Fail |
| Remarks | | | |
| Signature | | Quality assurance | |
| Verdict | ☐ Accepted ☐ Rejected | | |



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